

3 hours

Problem 1 (00IMO2)

We set $a = \frac{x}{y}$; $b = \frac{y}{z}$; $c = \frac{z}{x}$ for some positive x, y, z . We are showing that,

$$\prod_{cyc} \left(\frac{x}{y} - 1 + \frac{z}{y} \right) \leq 1$$
$$\Leftrightarrow \prod_{cyc} (x - y + z) \leq xyz$$

If any factor on the LHS is negative, then one of the variables is smaller than the other two. Therefore, at most one factor on the LHS may be negative. In this case, clearly negative LHS is smaller than positive RHS.

In the case where each of the factors on the LHS is positive, by AM-GM,

$$\sqrt{(x - y + z)(-x + y + z)} \leq x$$

Taking the product cyclically, we are done.

Problem 2 (10USATST2)

Using Holder's,

$$\left(\sum_{cyc} \frac{1}{a^5(b+2c)^2} \right) \left(\sum_{cyc} a(b+2c) \right)^2 \geq \left(\sum_{cyc} \frac{1}{a} \right)^3$$

It hence suffices to prove,

$$\frac{\left(\sum_{cyc} \frac{1}{a} \right)^3}{\left(\sum_{cyc} a(b+2c) \right)^2} \geq \frac{1}{3} \Leftrightarrow \left(\sum_{cyc} \frac{1}{a} \right)^3 \geq ab + bc + ca$$
$$\Leftrightarrow \left(\sum_{cyc} bc \right)^3 \geq ab + bc + ca \Leftrightarrow \left(\sum_{cyc} bc \right)^2 \geq 1$$
$$\Leftrightarrow (ab + bc + ca)^2 \geq a^{\frac{4}{3}} b^{\frac{4}{3}} c^{\frac{4}{3}}$$

This is AM-GM.

5 clubs; 4 hours

Problem 4 (18BMOSLA1)

We homogenize,

$$\sum_{cyc} \frac{1}{c(a+b)} \geq \frac{3}{2} \cdot \frac{a+b+c}{a^3+b^3+c^3}$$

WLOG, let $a+b+c=3$.

$$\sum_{cyc} \frac{1}{c(3-c)} \geq \frac{9}{2} \cdot \frac{1}{a^3+b^3+c^3}$$

Letting $f(x) = \frac{1}{x(3-x)}$ satisfies $f'(x) = \frac{2x-3}{x^2(3-x)^2}$ and,

$$f''(x) = \frac{2(x^2(3-x)^2) - (2x-3)(4x^3 - 18x^2 + 18x)}{x^4(3-x)^4}$$

We may check that this is non-negative on $x \in [0, 3]$ so f is convex on this range. Therefore, by Jensen's,

$$\sum_{cyc} \frac{1}{c(3-c)} \geq \frac{3}{2}$$

It suffices to show,

$$\begin{aligned} \frac{3}{2} &\geq \frac{9}{2} \cdot \frac{1}{a^3+b^3+c^3} \\ \Leftrightarrow a^3+b^3+c^3 &\geq \frac{(a+b+c)^3}{9} \end{aligned}$$

This is true by Power-Mean.

Problem 5 (01INMO3)

Approach 1

$$1 \leq a^a b^b c^c$$

Taking log of both sides as they are positive,

$$\Leftrightarrow 0 \leq \sum_{cyc} a \log(a)$$

We plug in $a = e^x; b = e^y; c = e^z$,

$$0 \leq \sum_{cyc} x e^x$$

We may finish with the tangent line trick.

Approach 2

$$a^{b+c} b^{c+a} c^{a+b} \leq 1$$

$$\Leftrightarrow \sqrt[2a+2b+2c]{a^{b+c} b^{c+a} c^{a+b}} \leq 1$$

As desired, by weighted AM-GM, [can we apply it even though exponents are variables? If so should we add a note on why it's possible]

$$\sqrt[2a+2b+2c]{a^{b+c} b^{c+a} c^{a+b}} \leq \frac{b+c}{2a+2b+2c} \cdot a + \frac{c+a}{2a+2b+2c} \cdot b + \frac{a+b}{2a+2b+2c} \cdot c = 1$$

10 clubs; 4 hours

Problem 6 (02MOP)

We claim that the answer is $S \in (2\sqrt{2}, \infty)$ for (i) and $S \in [3, \infty)$ for (ii).

Note that if we plug in $(x, 1, 1)$,

$$S = x^r + 2 \left(\frac{2}{1+x} \right)^r$$

Clearly, $f(x) = x^r + 2 \left(\frac{2}{1+x} \right)^r$ is continuous on positives. Note that, for both $r = \frac{1}{2}$ and $\frac{2}{3}$,

$$f(1) = 3; \lim_{x \rightarrow \infty} x^r + 2 \left(\frac{2}{1+x} \right)^r = +\infty$$

Therefore, any $S \in [3, \infty)$ is attainable for $r = \frac{2}{3}$ by the Intermediate Value Theorem.

Additionally, for $r = \frac{1}{2}$, we see that,

$$\lim_{x \rightarrow 0^+} S = 0 + 2(2)^{\frac{1}{2}} = 2\sqrt{2}$$

Therefore, any $S \in (2\sqrt{2}, \infty)$ is attainable also by the Intermediate Value Theorem. We now show no other values work, beginning with $r = \frac{2}{3}$. We show that,

$$\sum_{cyc} \left(\frac{2a}{b+c} \right)^{\frac{2}{3}} \geq 3$$

Note that the inequality is homogenous, so scale a, b, c such that $a + b + c = 3$.

$$\Leftrightarrow \sum_{cyc} \left(\frac{2a}{3-a} \right)^{\frac{2}{3}} \geq 3$$

Let $f(x) = \left(\frac{2x}{3-x} \right)^{\frac{2}{3}}$. We use the tangent trick,

$$f'(x) = \frac{2}{3} \left(\frac{2x}{3-x} \right)^{-\frac{1}{3}} \cdot \left(\frac{2(3-x) - 2x(-1)}{(3-x)^2} \right)$$

Plugging in $x = 1$, $f'(1) = \frac{2}{3}(1) \cdot \left(\frac{4+2}{4} \right) = 1$. We show that,

$$f(x) \geq f(1) + f'(1)(x-1)$$

$$\Leftrightarrow f(x) \geq 1 + (x-1) = x$$

$$\Leftrightarrow \left(\frac{2x}{3-x} \right)^{\frac{2}{3}} \geq x$$

On $x \in (0, 3)$, the domain of a, b, c , both the numerator and denominator of $\frac{2x}{3-x}$ are positive so we may do,

$$\Leftrightarrow 2x \geq (3-x)x^{\frac{3}{2}}$$

$$\Leftrightarrow 2 \geq (3-x)x^{\frac{1}{2}}$$

Plug in $y = x^{\frac{1}{2}}$,

$$\Leftrightarrow y^3 - 3y + 2 \geq 0$$

$$\Leftrightarrow (y-1)^2(y+2) \geq 0$$

This is clearly true, so we are done with (ii). We now do $r = \frac{1}{2}$,

$$S = \sum_{cyc} \left(\frac{2a}{3-a} \right)^{\frac{1}{2}} \geq 2\sqrt{2}$$

If we let $f(x) = \left(\frac{2x}{3-x} \right)^{\frac{1}{2}}$ and differentiate twice,

$$\begin{aligned} f'(x) &= \frac{1}{2} \left(\frac{2x}{3-x} \right)^{-\frac{1}{2}} \cdot \left(\frac{2(3-x) - 2x(-1)}{(3-x)^2} \right) \\ &= \frac{1}{2} \left(\frac{2x}{3-x} \right)^{-\frac{1}{2}} \cdot \left(\frac{6}{(3-x)^2} \right) \\ &= \frac{3\sqrt{2}}{2} \cdot x^{-\frac{1}{2}} (3-x)^{-\frac{3}{2}} \end{aligned}$$

Next,

$$\begin{aligned} f''(x) &= \frac{3\sqrt{2}}{2} \cdot \left(\left(-\frac{1}{2} \cdot x^{-\frac{3}{2}} \right) (3-x)^{-\frac{3}{2}} + \left(\frac{3}{2} (3-x)^{-\frac{5}{2}} \right) x^{-\frac{1}{2}} \right) \\ &= \frac{\frac{3\sqrt{2}}{2} \cdot \left(-\frac{1}{2} \cdot \frac{1}{x} + \frac{3}{2} \cdot \frac{1}{3-x} \right)}{x^{\frac{1}{2}} (3-x)^{\frac{3}{2}}} \end{aligned}$$

On $0 < x < 3$, since the denominator is positive,

$$\begin{aligned} f''(x) \geq 0 &\Leftrightarrow -\frac{1}{2} \cdot \frac{1}{x} + \frac{3}{2} \cdot \frac{1}{3-x} \geq 0 \\ &\Leftrightarrow \frac{3}{3-x} \geq \frac{1}{x} \\ &\Leftrightarrow 3x \geq 3-x \\ &\Leftrightarrow x \geq \frac{3}{4} \end{aligned}$$

Therefore, $f(x)$ is concave on $\left[0, \frac{3}{4}\right]$ and convex on $\left[\frac{3}{4}, 3\right)$.

Case 1: At least two of $a, b, c \in \left[\frac{3}{4}, 3\right)$.

We take two of $a, b, c \in \left[\frac{3}{4}, 3\right)$ and set both to their average. The sum $a + b + c = 3$ is still true.

By Jensen's, $\sum_{cyc} \left(\frac{2a}{3-a}\right)^{\frac{1}{2}}$ has not increased. We find the new sum $\sum_{cyc} \left(\frac{2a}{3-a}\right)^{\frac{1}{2}}$ and show it is at least $2\sqrt{2}$ to suffice. WLOG, let $b = a$. This means $c = 3 - 2a$ for $0 < a < \frac{3}{2}$,

$$\sum_{cyc} \left(\frac{2a}{3-a}\right)^{\frac{1}{2}} = 2 \left(\frac{2a}{3-a}\right)^{\frac{1}{2}} + \left(\frac{6-4a}{2a}\right)^{\frac{1}{2}} \geq 2\sqrt{2}$$

$$\Leftrightarrow 2 \left(\frac{2a}{3-a}\right)^{\frac{1}{2}} + \left(\frac{3-2a}{a}\right)^{\frac{1}{2}} \geq 2\sqrt{2}$$

Letting $g(a) = 2 \left(\frac{2a}{3-a}\right)^{\frac{1}{2}} + \left(\frac{3-2a}{a}\right)^{\frac{1}{2}}$ and differentiating g ,

$$\begin{aligned} g'(a) &= \left(\frac{2a}{3-a}\right)^{-\frac{1}{2}} \cdot \frac{2(3-a) - 2a(-1)}{(a-3)^2} + \frac{1}{2} \left(\frac{3-2a}{a}\right)^{-\frac{1}{2}} \cdot \frac{-2(a) - (3-2a)}{a^2} \\ &= \left(\frac{2a}{3-a}\right)^{-\frac{1}{2}} \cdot \frac{6}{(3-a)^2} - \frac{1}{2} \left(\frac{3-2a}{a}\right)^{-\frac{1}{2}} \cdot \frac{3}{a^2} \end{aligned}$$

For $g'(a) \geq 0$,

$$\begin{aligned} \left(\frac{2a}{3-a}\right)^{-\frac{1}{2}} \cdot \frac{6}{(3-a)^2} &\geq \frac{1}{2} \left(\frac{3-2a}{a}\right)^{-\frac{1}{2}} \cdot \frac{3}{a^2} \\ \Leftrightarrow \left(\frac{3-a}{2a}\right) \cdot \frac{36}{(3-a)^4} &\geq \frac{1}{4} \left(\frac{a}{3-2a}\right) \cdot \frac{9}{a^4} \\ \Leftrightarrow \frac{18}{a(3-a)^3} &\geq \frac{\frac{9}{4}}{a^3(3-2a)} \\ \Leftrightarrow 8a^2(3-2a) &\geq s(3-a)^3 \\ \Leftrightarrow 0 \leq 15a^3 - 15a^2 - 27a + 27 \\ \Leftrightarrow 0 \leq (a-1)(15a^2 - 27) \end{aligned}$$

g is optimized at values where $g'(a)$ or at endpoints. Plugging $a = 1$, we see that $g(1) = 3$.

Since the derivative is possible from $a = 1$ to $a = \frac{3}{\sqrt{5}}$, $g\left(\frac{3}{\sqrt{5}}\right) > g(1)$. It then remains to check the endpoints. Since the limits to zero and $\frac{3}{2}$ are ∞ and $2\sqrt{2}$ respectively, but the function never reaches these limits, we are done.

Case 2: At least two of $a, b, c \in \left[0, \frac{3}{4}\right]$.

By Pigeonhole, it is impossible for all three of a, b, c to be on $\left[0, \frac{3}{4}\right]$. The only case that remains is where two of them are on this range (WLOG, say, a and b). If $a + b \leq \frac{3}{4}$, we may decrease a to zero and increase b to $a + b$. The sum $a + b + c = 3$ is still true while the total $\sum_{cyc} \left(\frac{2a}{3-a}\right)^{\frac{1}{2}}$ has decreased by Karamata. In this state, $a = 0$, and by AM-GM, as desired,

$$S = \left(\frac{2b}{c}\right)^{\frac{1}{2}} + \left(\frac{2c}{b}\right)^{\frac{1}{2}} \geq 2\sqrt{2}$$

Therefore, prior to the smoothing and before $\sum_{cyc} \left(\frac{2a}{3-a}\right)^{\frac{1}{2}}$ decreased, we had $\sum_{cyc} \left(\frac{2a}{3-a}\right)^{\frac{1}{2}} > 2\sqrt{2}$.

Problem 7 (12IMO2)

Note that, by weighted AM-GM,

$$\begin{aligned} \left(\frac{1 \cdot 1 + 1 \cdot a_2}{2}\right)^2 &\geq a_2; \\ \left(\frac{2 \cdot \left(\frac{1}{2}\right) + 1 \cdot a_3}{3}\right)^3 &\geq \left(\frac{1}{2}\right)^2 a_3; \\ &\dots \\ \left(\frac{(n-1) \cdot \left(\frac{1}{n-1}\right) + 1 \cdot a_n}{n}\right)^n &\geq \left(\frac{1}{n-1}\right)^{n-1} a_n \end{aligned}$$

Taking the product, we have,

$$\begin{aligned} (1 + a_2)^2 (1 + a_3)^3 \dots (1 + a_n)^n \left(\frac{1}{2}\right)^2 \dots \left(\frac{1}{n}\right)^n &\geq a_2 a_3 \dots a_n \left(\frac{1}{2}\right)^2 \dots \left(\frac{1}{n-1}\right)^{n-1} \\ \Leftrightarrow (1 + a_2)^2 (1 + a_3)^3 \dots (1 + a_n)^n &\geq n^n \end{aligned}$$

However, the equality case holds if and only if the equality case for each of the earlier equations hold. This would require $a_i = \frac{1}{i-1}$ which does not satisfy $\prod a_i = 1$.

16 clubs; 7 hours

Problem 9 (03ELMO4)

Main Claim: Given $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1$, we have $\sum_{cyc} \frac{1}{x^2-1} \geq 1$ for $x, y, z > 1$.

Proof: Let $a = \frac{1}{x+1}$; $b = \frac{1}{y+1}$; $c = \frac{1}{z+1}$. Note that $a + b + c = 1$ for some $a, b, c < \frac{1}{2}$ and we're trying to prove,

$$\sum_{cyc} \frac{1}{x^2-1} \geq 1 \Leftrightarrow \sum_{cyc} \frac{a^2}{1-2a} \geq 1$$

Indeed, if we let $f(x) = \frac{x^2}{1-2x}$, we have,

$$\begin{aligned} \Rightarrow f'(x) &= \frac{2x(1-2x) - x^2(-2)}{(1-2x)^2} = \frac{2x - 2x^2}{(1-2x)^2} \\ \Rightarrow f''(x) &= \frac{(2-4x)(1-2x)^2 - (2x-2x^2)(-4)(1-2x)}{(1-2x)^4} \\ &= \frac{(2-4x)(1-2x) + (8x-8x^2)}{(1-2x)^3} = \frac{2}{(1-2x)^3} \end{aligned}$$

Note that $f''(x) > 0$ for $x < \frac{1}{2}$ meaning f is convex on the domain of a, b, c . We are done by Jensen's. $\square$

With the main claim shown, FTSOC, let $\sum_{cyc} \frac{1}{x^2-1} = 1$ and $\sum_{cyc} \frac{1}{x+1} > 1$. We may increase x, y, z by some amount t until $\sum_{cyc} \frac{1}{x+1} = 1$ (this is possible by the Intermediate Value Theorem since setting t arbitrarily large will have $\sum_{cyc} \frac{1}{x+1}$ approach zero). Doing this must have caused $\sum_{cyc} \frac{1}{x^2-1}$ to decrease. Therefore, we'd have a triplet (x, y, z) such that $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1$ and $\sum_{cyc} \frac{1}{x^2-1} < 1$ which we've shown is impossible.

Problem 11 (06SLA5)

Let $x = \sqrt{b} + \sqrt{c} - \sqrt{a}$ and define y, z analogously.

Claim: x, y, z are positive.

Proof:

$$\begin{aligned}
& \sqrt{b} + \sqrt{c} - \sqrt{a} > 0 \\
& \Leftrightarrow \sqrt{b} + \sqrt{c} > \sqrt{a} \\
& \Leftrightarrow b + c + 2\sqrt{bc} > a \\
& \Leftrightarrow b + c > a
\end{aligned}$$

Analogously, y and z are positive as well. $\square$

Note that, $y + z = 2\sqrt{a} \Leftrightarrow a = \left(\frac{y+z}{2}\right)^2$. We may similarly write b, c in terms of x, y, z . Therefore,

$$\begin{aligned}
& \sum_{cyc} \frac{\sqrt{b+c-a}}{\sqrt{b} + \sqrt{c} - \sqrt{a}} \leq 3 \\
& \Leftrightarrow \sum_{cyc} \frac{\sqrt{\frac{(x+z)^2}{4} + \frac{(x+y)^2}{4} - \frac{(y+z)^2}{4}}}{x} \leq 3 \\
& \Leftrightarrow \sum_{cyc} \frac{\sqrt{2x^2 + 2xz + 2xy - 2yz}}{2x} \leq 3 \\
& \Leftrightarrow \frac{\sum_{cyc} \frac{\sqrt{2x^2 + 2xz + 2xy - 2yz}}{2x}}{3} \leq 1
\end{aligned}$$

Note that, by AM-QM,

$$\frac{\sum_{cyc} \frac{\sqrt{2x^2 + 2xz + 2xy - 2yz}}{2x}}{3} \leq \sqrt{\frac{\sum_{cyc} \frac{1}{2} + \frac{z}{2x} + \frac{y}{2x} - \frac{yz}{2x^2}}{3}}$$

Therefore, it suffices to prove that,

$$\begin{aligned}
& \sqrt{\frac{\sum_{cyc} \frac{1}{2} + \frac{z}{2x} + \frac{y}{2x} - \frac{yz}{2x^2}}{3}} \leq 1 \\
& \Leftrightarrow \sum_{cyc} \frac{1}{2} + \frac{z}{2x} + \frac{y}{2x} - \frac{yz}{2x^2} \leq 3 \\
& \Leftrightarrow \sum_{cyc} \frac{z}{2x} + \frac{y}{2x} \leq \sum_{cyc} \frac{1}{2} + \frac{yz}{2x^2}
\end{aligned}$$

$$\Leftrightarrow \sum_{cyc} xy^2z^3 + xy^3z^2 \leq \sum_{cyc} x^2y^2z^2 + y^3z^3$$

We substitute $m = yz; n = zx; k = xy$,

$$\Leftrightarrow \sum_{cyc} nm^2 + km^2 \leq \sum_{cyc} mnk + m^3$$

This is Schur's inequality.

22 clubs; 9 hours

Problem 14 (11MOPR42)

We rewrite the RHS,

$$\begin{aligned} & \sqrt{\frac{a^3 + b^3}{a + b}} + \sqrt{\frac{b^3 + c^3}{b + c}} + \sqrt{\frac{c^3 + a^3}{c + a}} + 9\sqrt[3]{abc} \leq 4(a + b + c) \\ & \Leftrightarrow \sum_{cyc} \sqrt{a^2 - ab + b^2} + 9\sqrt[3]{abc} \leq 4(a + b + c) \end{aligned}$$

By AM-GM, $9\sqrt[3]{abc} \leq 3(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})$. Therefore,

$$\Leftrightarrow \sum_{cyc} (\sqrt{a^2 - ab + b^2} + 3\sqrt{ab}) \leq 4(a + b + c)$$

Indeed, by weighted AM-QM, $\frac{\sqrt{a^2 - ab + b^2} + 3\sqrt{ab}}{4} \leq \sqrt{\frac{a^2 + 2ab + b^2}{4}} = \frac{a + b}{2}$.

$$\Leftrightarrow \sum_{cyc} 2(a + b) \leq 4(a + b + c)$$

This is an equality, so we are done.

Problem 15 (17AMO6)

We claim the minimum is $\frac{2}{3}$ attained with $(2, 2, 0, 0)$.

Taking the tangent line trick at two on $f(x) = \frac{1}{x^3 + 4}$, we prove the following.

Claim: $\frac{1}{x^3 + 4} \geq -\frac{1}{12}x + \frac{1}{4}$ for all non-negative x .

Proof:

$$\begin{aligned}\frac{1}{x^3 + 4} &\geq -\frac{1}{12}x + \frac{1}{4} \\ \Leftrightarrow -12 &\leq x^4 + 4x - 3x^3 - 12 \\ \Leftrightarrow 0 &\geq x(x+1)(x-2)^2\end{aligned}$$

This is clearly true for non-negative x . $\square$

Therefore,

$$\begin{aligned}\sum_{cyc} \frac{a}{b^3 + 4} &\geq \frac{2}{3} \\ \Leftrightarrow \sum_{cyc} \frac{a}{4} - \frac{ab}{12} &\geq \frac{2}{3} \Leftrightarrow \sum_{cyc} -\frac{ab}{12} \geq -\frac{1}{3} \\ \Leftrightarrow \sum_{cyc} ab &\leq 4 \Leftrightarrow (a+c)(b+d) \leq 4\end{aligned}$$

This is true by AM-GM.

32 clubs; 12 hours

Problem 16 (18AMO1)

Everything is homogenous so WLOG $a + b + c = 4 \Leftrightarrow abc = 1$.

$$\begin{aligned}2(ab + bc + ca) + 4 \min(a^2, b^2, c^2) &\geq a^2 + b^2 + c^2 \\ \Leftrightarrow 4(ab + bc + ca) + 4 \min(a^2, b^2, c^2) &\geq (a + b + c)^2 \\ \Leftrightarrow (ab + bc + ca) + \min(a^2, b^2, c^2) &\geq 4\end{aligned}$$

WLOG, let $a \leq b \leq c$,

$$\Leftrightarrow ab + bc + ca + a^2 \geq 4 \Leftrightarrow (a + b + c)a + bc \geq 4 \Leftrightarrow 4a + \frac{1}{a} \geq 4$$

We are done with AM-GM.

Problem 19 (20IMO2)

By weighted AM-GM, we know,

$$\begin{aligned} \sqrt[a+b+c+d]{a^a b^b c^c d^d} &< \frac{a \cdot a + b \cdot b + c \cdot c + d \cdot d}{a + b + c + d} \\ \Leftrightarrow a^a b^b c^c d^d &< a^2 + b^2 + c^2 + d^2 \end{aligned}$$

Therefore,

$$\begin{aligned} (a + 2b + 3c + 4d)a^a b^b c^c d^d &< 1 \\ \Leftrightarrow (a + 2b + 3c + 4d)(a^2 + b^2 + c^2 + d^2) &< (a + b + c + d)^3 \\ \Leftrightarrow a^3 + 2b^3 + 3c^3 + 4d^3 + a(b^2 + c^2 + d^2) + \dots + 4d(a^2 + b^2 + c^2) \\ &< a^3 + b^3 + c^3 + d^3 + \sum_{cyc} 3a(b^2 + c^2 + d^2) + \sum_{sym} abc \\ \Leftrightarrow b^3 + 2c^3 + 3d^2 + d(a^2 + b^2 + c^2) &< 2a(b^2 + c^2 + d^2) + b(a^2 + c^2 + d^2) + \sum_{sym} abc \end{aligned}$$

Indeed, we have $a \geq b \geq c \geq d > 0$ therefore $b^3 \leq ab^2$; $2c^3 \leq 2ac^2$; $3d^2 \leq 2ad^2 + bd^2$; $d(a^2 + b^2 + c^2) \leq ab^2 + ab^2 + bc^2$. The remaining $0 < \sum_{sym} abc$ as everything is positive. Adding these up gives the desired result.

40 clubs; 13.5 hours

Problem 21 (05IMO3)

$$\begin{aligned} \sum_{cyc} \frac{x^5 - x^2}{x^5 + y^2 + z^2} &\geq 0 \\ \Leftrightarrow \sum_{cyc} \frac{-x^2 - y^2 - z^2}{x^5 + y^2 + z^2} &\geq -3 \\ \Leftrightarrow \sum_{cyc} \frac{1}{x^5 + y^2 + z^2} &\leq \frac{3}{x^2 + y^2 + z^2} \end{aligned}$$

Note that, if $xyz > 1$, then we may scale x, y, z by the same $t < 1$ such that the numerators $x^2 + y^2 + z^2$ will each be scaled by t^2 but the denominators $x^5 + y^2 + z^2$ will be scaled by a factor strictly between t^2 and t^5 . This would hence increase each fraction $\frac{x^2 + y^2 + z^2}{x^5 + y^2 + z^2}$.

Therefore, we may assume $xyz = 1$. [Should/can the wording in this paragraph be improved?]

However, by Cauchy-Schwarz,

$$\begin{aligned}
& (x^5 + y^2 + z^2) \left(\frac{1}{x} + y^2 + z^2 \right) \geq (x^2 + y^2 + z^2)^2 \\
& \Leftrightarrow (x^5 + y^2 + z^2)(yz + y^2 + z^2) \geq (x^2 + y^2 + z^2)^2 \\
& \Leftrightarrow \frac{x^2 + y^2 + z^2}{x^5 + y^2 + z^2} \leq \frac{yz + y^2 + z^2}{x^2 + y^2 + z^2}
\end{aligned}$$

Therefore, it suffices to show,

$$\begin{aligned}
& \sum_{cyc} \frac{yz + y^2 + z^2}{x^2 + y^2 + z^2} \leq 3 \\
& \Leftrightarrow 2 \sum_{cyc} x^2 + \sum_{cyc} yz \leq 3 \sum_{cyc} x^2
\end{aligned}$$

This is clearly true by Muirhead.

Problem 24 (16SLA1)

Lemma: Given $ab \geq 1$, $(a^2 + 1)(b^2 + 1) \leq \left(\left(\frac{a+b}{2} \right)^2 + 1 \right)^2$.

Proof: Expanding and factor.

With the lemma proven,

$$\begin{aligned}
& \sqrt[3]{(a^2 + 1)(b^2 + 1)(c^2 + 1)} \leq \left(\frac{a + b + c}{3} \right)^2 + 1 \\
& \Leftrightarrow \sqrt[3]{\left(\left(\frac{a+b}{2} \right)^2 + 1 \right)^2 (c^2 + 1)} \leq \left(\frac{a + b + c}{3} \right)^2 + 1
\end{aligned}$$

We may take log of both sides,

$$\Leftrightarrow \frac{2 \log \left(\left(\frac{a+b}{2} \right)^2 + 1 \right) + \log(c^2 + 1)}{3} \leq \log \left(\left(\frac{a + b + c}{3} \right)^2 + 1 \right)$$

If we let $f(a) = \log(a^2 + 1)$,

$$\Leftrightarrow \frac{2f\left(\frac{a+b}{2}\right) + f(c)}{3} \leq f\left(\left(\frac{2\left(\frac{a+b}{2}\right) + c}{3}\right)^2 + 1\right)$$

Note that f is concave on $[1, \infty)$ so we are done.

50 clubs; 17 hours